Quantification: I534_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	13
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2941.54	2951	1607.61	px^2
major_axis_length_px	lacuna shape	85.8598	83.1379	30.1528	px
minor_axis_length_px	lacuna shape	45.6621	45.1555	14.9313	px
aspect_ratio	lacuna shape	1.9335	1.9125	0.5477	unitless
eccentricity	lacuna shape	0.8143	0.8524	0.1082	unitless
solidity	lacuna shape	0.8338	0.8509	0.098	unitless
roots_count	headline	8.7692	7	4.106	count
ring_length_r30_px	headline	288.846	267	118.711	px
ring_length_r60_px	reported	620.846	607	191.075	px
owned_length_px	ownership-dependent	1067.97	882.583	510.786	px
edge_count	ownership-dependent	50.3077	50	23.4428	count
mean_edge_length_px	ownership-dependent	21.1512	21.2502	2.8912	px
perimeter_px	normalised	273.55	262.877	100.444	px
ring_area_r30_px2	normalised	9806.31	9533	2617.99	px^2
ring_area_r60_px2	normalised	22647.7	22663	5117.86	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.99	1	0.0267	unitless
ring_density_r30	normalised	0.0289937	0.0252998	0.00700484	px^-1
ring_density_r60	normalised	0.0274083	0.0256471	0.00598118	px^-1
roots_per_100px_perimeter	normalised	3.2104	3.2553	0.6594	per 100 px
ring_attached_length_r30_px	network v2	260.923	249	118.419	px
ring_attached_length_r60_px	network v2	486.692	503	198.802	px
ring_length_w_r30_px	network v2	328.987	302.936	136.316	px
ring_length_w_r60_px	network v2	709.597	690.47	215.166	px
sholl_crossings_r10	network v2	6.6923	7	2.4285	count
sholl_crossings_r20	network v2	7.8462	6	3.9125	count
sholl_crossings_r30	network v2	9	8	3.2404	count
loop_count	loops, report only	2.5385	2	2.4364	count
loop_edge_share	loops, report only	0.1643	0.1667	0.0979	share
ring_width_mean_r30_px	width	7.3466	7.3202	0.5047	px

Per field
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measure	value	units
analysed_area_px2	877616	px^2
total_skeleton_length_px	22676	px
canalicular_length_density_per_px	0.0258382	px^-1
junction_count	456	count
junction_density_per_px2	0.00051959	px^-2
lacuna_count	17	count

lacunae_per_px2	1.937e-05	px^-2
skeleton_component_count	297	count
mean_component_length_px	76.3502	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0293825	px^-1
field_density_without_flagged_per_px	0.0228044	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	175	px
unattached_component_count	12	count
canalicular_length_density_attached_per_px	0.0258382	px^-1
field_length_density_with_unattached_per_px	0.0260376	px^-1
excluded_area_px2	127816	px^2
analysed_field_area_px2	920760	px^2
loop_count	42	count
loop_edge_share	0.1153	share
edge_count	1249	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9044	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	22410	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
2	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
3	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
4	749	51.0208	20.659	2.4697	0.9144	0.8874	1.1423	366.493	18.0734	True
5	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
6	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
7	623	33.6165	26.9149	1.249	0.5991	0.8534	-1.4105	448.144	1010.89	True
8	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
9	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
10	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
11	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
12	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
13	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
14	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False
15	812	45.0748	28.6241	1.5747	0.7725	0.7785	0.6665	738.431	13.2463	True
16	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
17	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

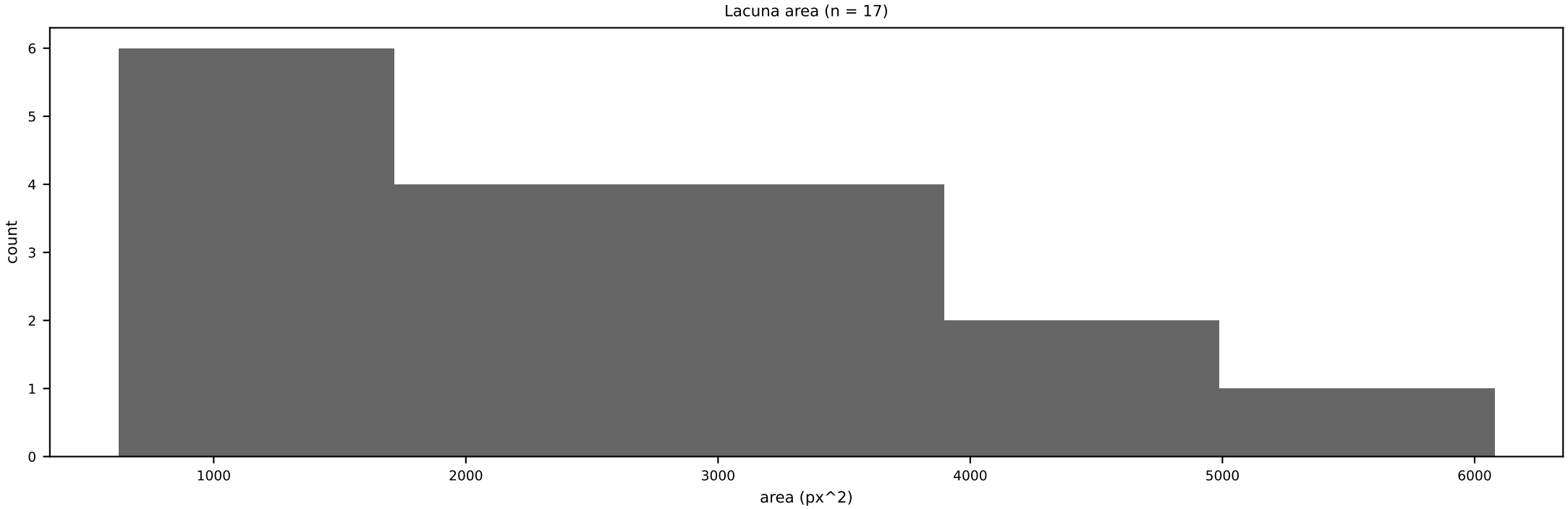
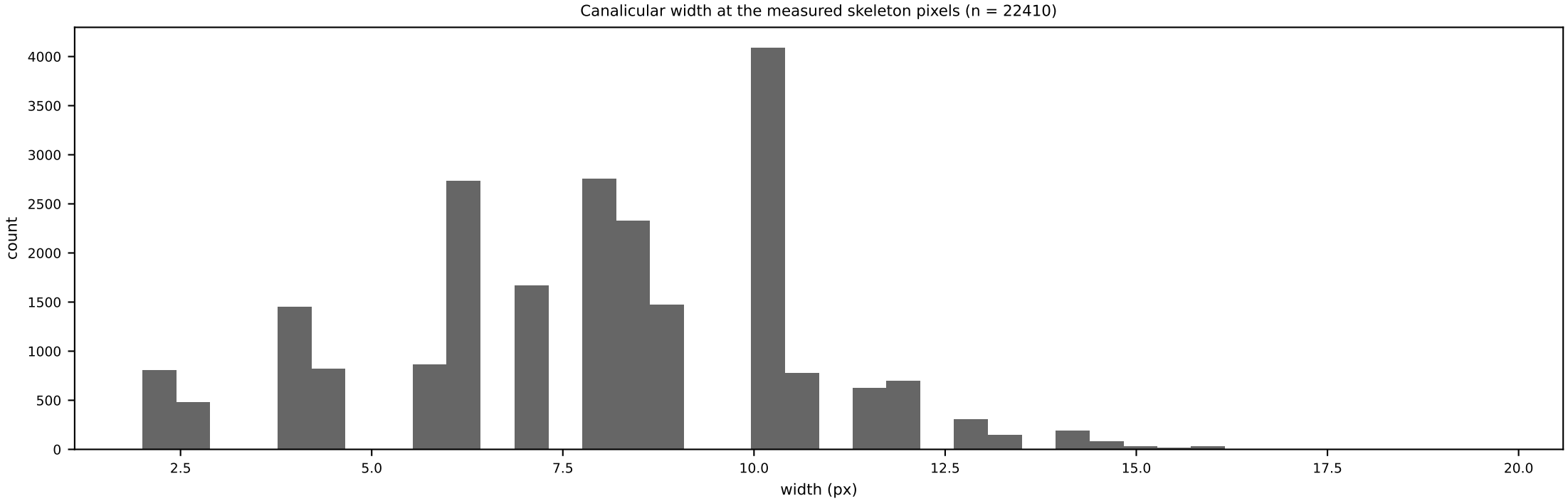
1	5	267	678	2052.44	106	19.3626	7.814
2	5	67	216	348.191	17	20.4818	5.652
3	7	282	607	1224.49	55	22.2634	6.9732
4	3	126	307	183.61	6	30.6017	6.8573
5	13	460	1042	1642.25	78	21.0545	7.061
6	12	359	545	737.097	43	17.1418	7.3776
7	4	130	342	346.635	13	26.6642	7.5691
8	7	218	548	878.826	42	20.9244	8.3837
9	6	215	640	517.546	29	17.8464	7.2371
10	18	511	844	882.583	51	17.3055	7.3445
11	8	300	687	1332.88	50	26.6576	7.3202
12	14	434	758	1129.97	52	21.7303	7.5424
13	6	202	494	1734.65	67	25.8902	7.9784
14	6	142	292	322.635	15	21.509	7.2157
15	6	80	152	128.054	9	14.2282	5.2875
16	7	211	437	701.257	33	21.2502	6.7776
17	5	154	499	726.968	33	22.0293	6.4806

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	132.125	6322	18207	1	1	0.0422335	0.0372384	3.7843
2	237.965	7174	17084	0.7631	0.6986	0.00933928	0.0126434	2.1011
3	334.475	11891	27068	1	1	0.0237154	0.022425	2.0928
4	128.947	4467	9758	0.694	0.6341	0.0282068	0.0314614	2.3265
5	419.345	11919	24709	1	1	0.0385938	0.0421709	3.1001
6	332.676	11820	24257	1	1	0.0303722	0.0224677	3.6071
7	114.184	3796	10383	0.6248	0.5844	0.0342466	0.0329385	3.5031
8	249.078	9533	20117	1	0.9067	0.0228679	0.0272406	2.8104
9	262.877	9914	25300	1	1	0.0216865	0.0252964	2.2824
10	405.588	13364	31075	1	1	0.0382371	0.0271601	4.438
11	278.09	9508	22663	1	1	0.0315524	0.0303137	2.8768
12	406.416	13768	29555	1	1	0.0315224	0.0256471	3.4447
13	220.35	8451	20862	1	1	0.0239025	0.0236794	2.7229
14	153.238	5825	12202	1	0.9659	0.0243777	0.0239305	3.9155
15	188.368	3275	4996	0.6244	0.5829	0.0244275	0.0304243	3.1853
16	215.037	8340	19166	1	0.9972	0.0252998	0.0228008	3.2553
17	146.853	6827	19239	1	1	0.0225575	0.0259369	3.4048

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	253	550	302.936	780.725	7	6	9
2	67	67	79.2548	247.551	2	1	0
3	249	503	315.451	690.47	7	5	8
4	114	152	139.74	352.806	3	2	6
5	416	877	530.345	1185.18	10	16	16
6	349	535	418.161	626.357	6	10	9
7	109	303	145.811	390.605	2	3	2
8	179	412	245.409	621.044	5	9	8
9	120	199	234.024	722.227	3	5	8
10	497	735	579.902	953.98	11	14	9
11	270	583	338.007	796.252	7	9	11
12	390	664	496.375	859.039	10	10	14
13	177	309	235.38	579.399	5	6	7
14	141	246	163.61	338.049	4	4	7
15	64	64	84.1127	169.953	1	0	2

16	200	330	235.752	490.161	7	4	8
17	151	384	181.48	581.884	5	4	3

I534_z25c1-2: distributions, pre-validation, px